

Raspberry pi and Python for IoT

Agenda

- Operating systems for Rpi
- General Purpose Input Output (GPIO)
- Schematic Circuit Editor
- Send out DATA from a RPi

Operating systems for RPi

- Many operating systems are available for Raspberry Pi that are categorized based on their features and applications.
- RPi OS with desktop
- RPi OS Lite
- RPi OS Server
- RPi OS for specific application



ubuntu **MATE**

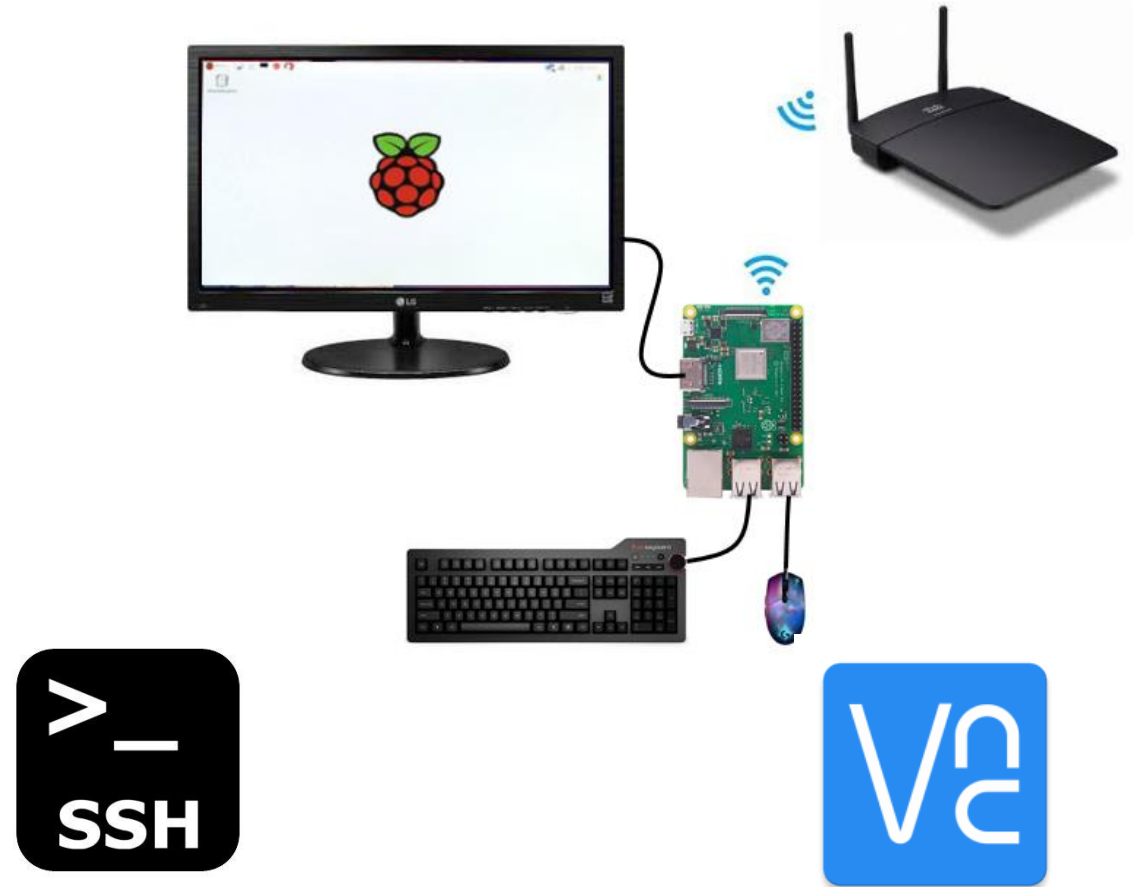
OS Installation Requirements

- OS image
- memory stick
- Memory card reader
- OS imager application



Connect to a RPi

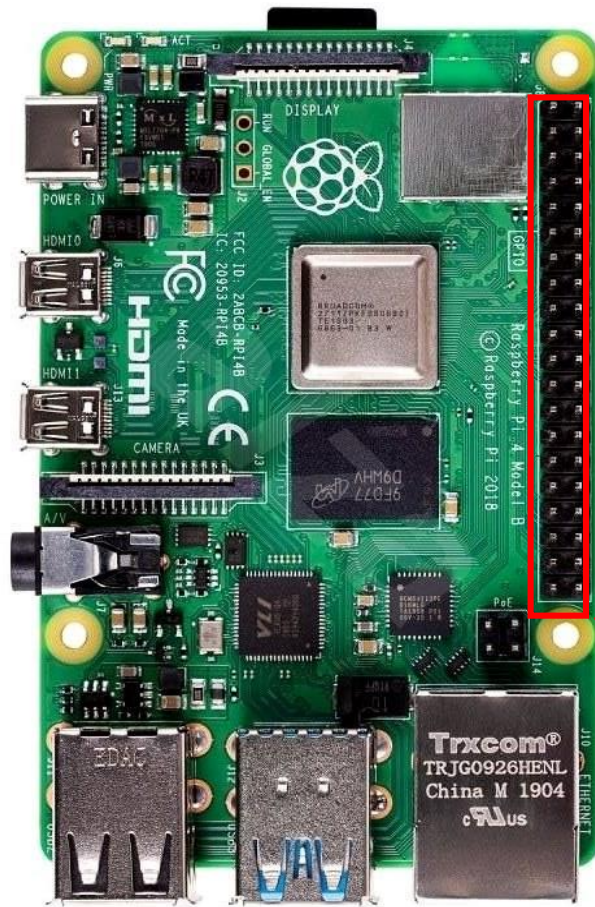
- A network connection
- RPi IP address
- Remote connection service
 - SSH
 - VNC



General Purpose Input/Output (GPIO)

- GPIO is an interface available on most modern SBCs and microcontrollers to provide an ease of access to the devices internal properties.
- Each GPIO should be able to define either an input mode or an output mode for individual pins on the chip.
- There are 40 pins on the Raspberry Pi (26 pins on early models), and they provide various different functions. These pins are a physical interface between the Raspberry Pi and the outside world.

RPi GPIO Pins



3v3 Power	1	5v Power
GPIO 2 (I2C1 SDA)	3	5v Power
GPIO 3 (I2C1 SCL)	5	Ground
GPIO 4 (GPCLK0)	7	GPIO 14 (UART TX)
Ground	9	GPIO 15 (UART RX)
GPIO 17	11	GPIO 18 (PCM CLK)
GPIO 27	13	Ground
GPIO 22	15	GPIO 23
3v3 Power	17	GPIO 24
GPIO 10 (SPI0 MOSI)	19	Ground
GPIO 9 (SPI0 MISO)	21	GPIO 25
GPIO 11 (SPI0 SCLK)	23	GPIO 8 (SPI0 CE0)
Ground	25	GPIO 7 (SPI0 CE1)
GPIO 0 (EEPROM SDA)	27	GPIO 1 (EEPROM SCL)
GPIO 5	29	Ground
GPIO 6	31	GPIO 12 (PWM0)
GPIO 13 (PWM1)	33	Ground
GPIO 19 (PCM FS)	35	GPIO 16
GPIO 26	37	GPIO 20 (PCM DIN)
Ground	39	GPIO 21 (PCM DOUT)

- GPIO (General Purpose IO)
- SPI (Serial Peripheral Interface)
- I²C (Inter-integrated Circuit)
- UART (Universal Asynchronous Receiver/Transmitter)
- PCM (Pulse Code Modulation)
- Ground
- 5v (Power)
- 3.3v (Power)

RPi GPIO in Python

- Python has powerful modules (e.g., RPi.GPIO) to control the GPIO interface on the Raspberry Pi.
- RPi GPIO has two modes: BCM and Board.
 - BCM stands for Broadcom SOC channel that refers to Raspberry Pi's Broadcom-chip brain.
 - Board refers to the physical number of pins

1. *import RPi.GPIO as GPIO*
2. *GPIO.setmode(GPIO.BCM)*
3. *GPIO.setmode(GPIO.BOARD)*

GPIO Port Functionality

After setting the board mode, the direction and function of each port should be selected.

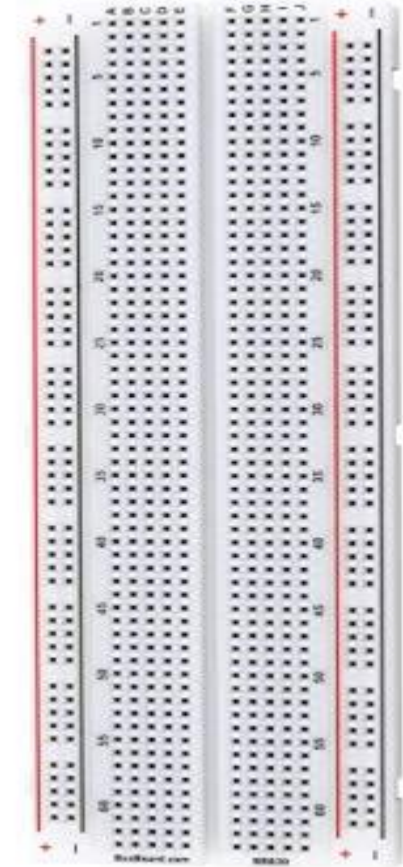
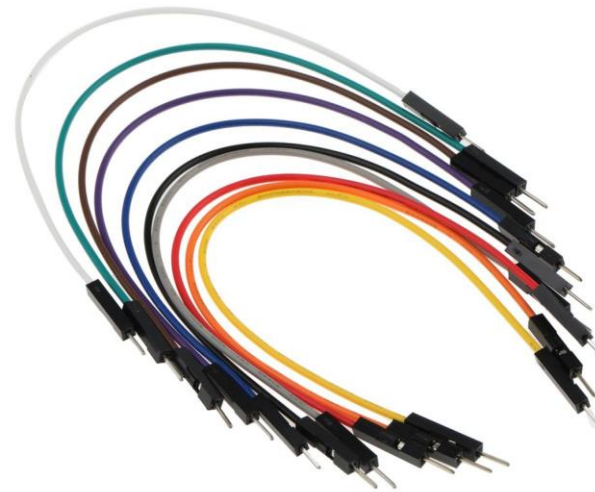
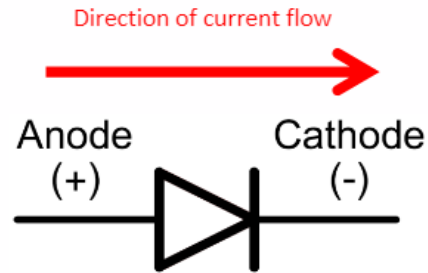
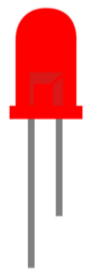
GPIO.setup(port_or_pin, GPIO.IN) # set port/pin as an input

GPIO.setup(port_or_pin, GPIO.OUT) # set port/pin as an output

GPIO.output(port_or_pin, 1) # set an output port/pin value to 1 / GPIO.HIGH / True

Send out DATA from a RPi

Required physical components:



Schematic Circuit Editor

- Circuit diagram is a visual representation of an electrical circuit that is made using basic images of parts used or even symbols that are used in the industry.
- Schematic version of circuit makers uses the industry level symbol.

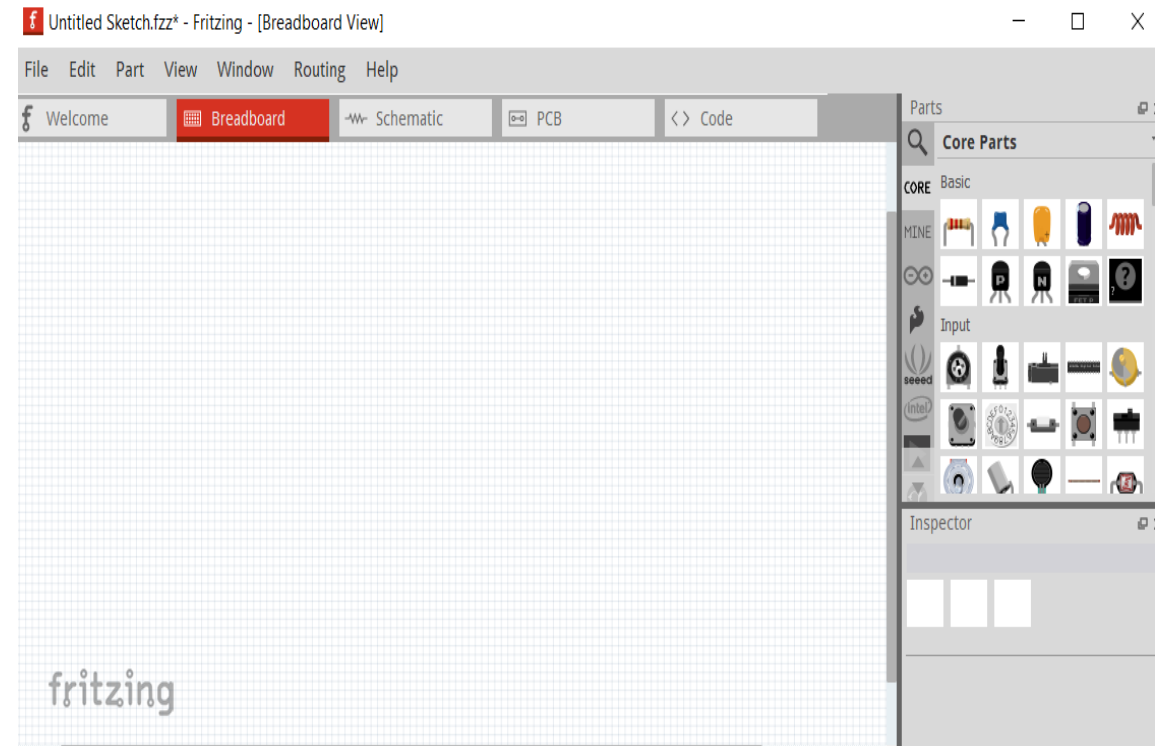


Fritzing

- Fritzing is an open-source hardware initiative that makes electronics accessible as a creative material for anyone.
- Fritzing is open-source EDA (Electronic Design Automation) software for people who are not engineers.

<https://fritzing.org/>

<https://www.youtube.com/watch?v=Hxhd4HKrWpg>



Electrical resistance

Electrical resistance is opposition to the flow of current in an electric circuit: resistance converts electrical energy to thermal energy.

For many materials, electrical resistance, R , is defined by Ohm's law:

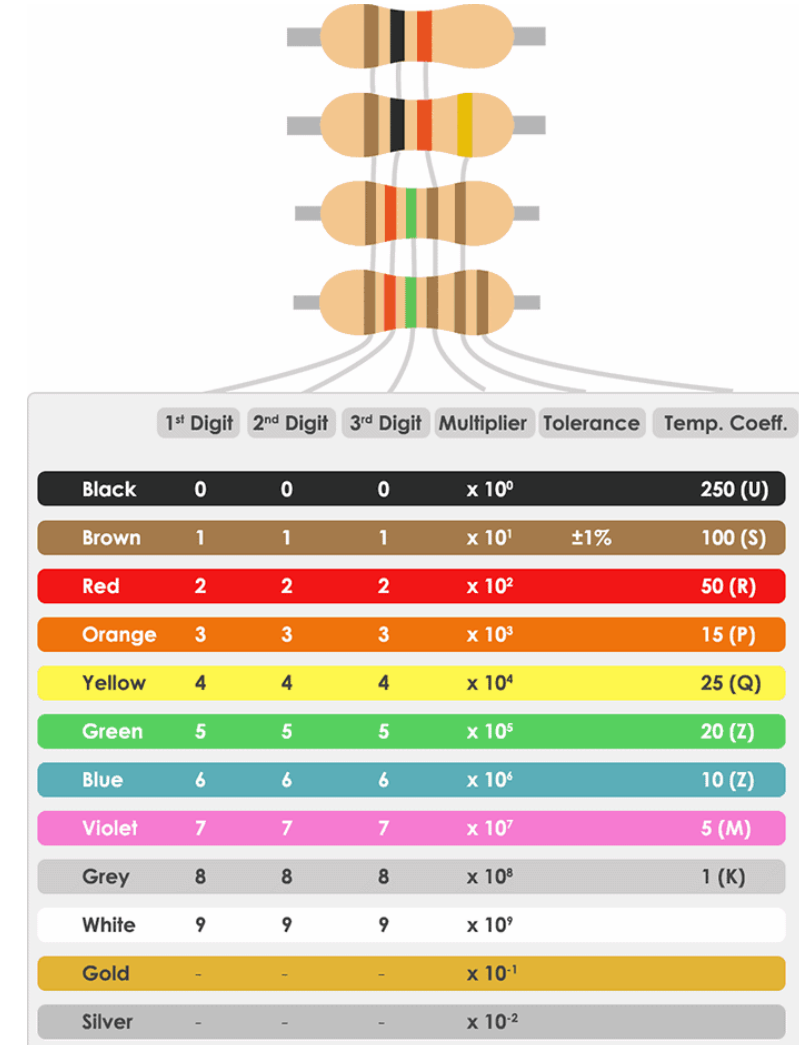
$$R = V / I$$

where V is voltage in volts and I is current in amps. The unit of resistance is the ohm, symbol Ω .



The Resistor Color Code

- The code is a convenient way to tell the resistance of a resistor.
- Colored bands are placed on a resistor. Each band has a numerical value
- Bands are read from left to right

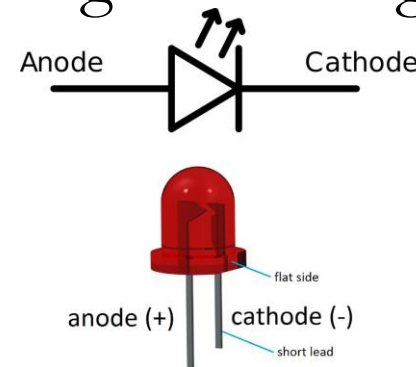
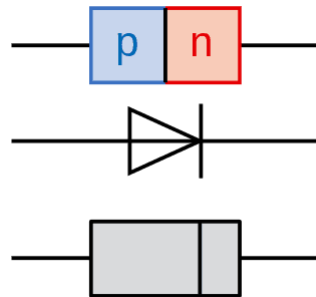


	1 st Digit	2 nd Digit	3 rd Digit	Multiplier	Tolerance	Temp. Coeff.
Black	0	0	0	$\times 10^0$		250 (U)
Brown	1	1	1	$\times 10^1$	$\pm 1\%$	100 (S)
Red	2	2	2	$\times 10^2$		50 (R)
Orange	3	3	3	$\times 10^3$		15 (P)
Yellow	4	4	4	$\times 10^4$		25 (Q)
Green	5	5	5	$\times 10^5$		20 (Z)
Blue	6	6	6	$\times 10^6$		10 (Z)
Violet	7	7	7	$\times 10^7$		5 (M)
Grey	8	8	8	$\times 10^8$		1 (K)
White	9	9	9	$\times 10^9$		
Gold	-	-	-	$\times 10^{-1}$		
Silver	-	-	-	$\times 10^{-2}$		

<https://www.calculator.net/resistor-calculator.html>

Light Emitting Diode (**LED**)

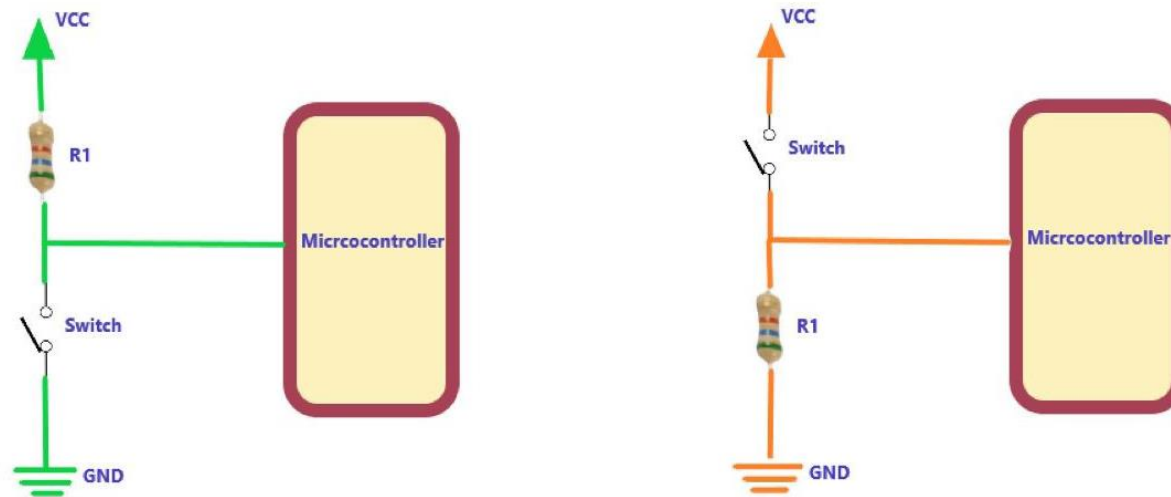
- A diode is a semiconductor device that essentially acts as a one-way switch for current.
- Diodes have polarity, determined by an anode (positive lead) and cathode (negative lead). Most diodes allow current to flow only when positive voltage is applied to the anode.
- All diodes emit photons (particles of electromagnetic energy), but only certain types of diodes emit that electromagnetic energy as light instead of heat.



GPIO Port Functionality

`GPIO.setup(port_or_pin, GPIO.IN, pull_up_down=GPIO.PUD_DOWN) # input with pull-down`

`GPIO.setup(port_or_pin, GPIO.IN, pull_up_down=GPIO.PUD_UP) # input with pull-up`



LED with RPi Circuit

